

Sunny Zhang

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Education

University of Toronto, St. George

Honours BSc, Double Major in Computer Science and Mathematics

Expected 2029

GPA: 3.7/4.0

Algoverse AI Research Program

Jun 2026 – Present

Selected research cohort conducting applied AI/ML research under mentor guidance.

Technical Skills

Languages: Python, C, C++, Java, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks/Libraries: Next.js, React, Tailwind CSS, FastAPI, Flask, SQLAlchemy, Pandas, NumPy, PyTorch, TensorFlow, scikit-learn, D3

Tools/Platforms: Git, Docker, PostgreSQL, SQLite, Supabase, Auth0, Gemini API, pytest, Vercel

Projects

Tiny Society AI | [Live Demo](#) · [GitHub](#)

Next.js, FastAPI, Python, Supabase, D3

- Engineered a full-stack multi-agent social simulation and prediction engine where 6+ AI agents reason, remember, reflect, form relationships, and shift opinions over 40+ simulated days.
- Implemented relevance-weighted memory retrieval (semantic relevance, exponential recency decay, and importance scoring) with reflection and goal-driven planning, following the Stanford Generative-Agents architecture.
- Designed a forecasting layer that aggregates per-agent beliefs into population-level topic distributions, uncertainty scores, confidence estimates, and pivotal-day traces.
- Streamed live simulations over Server-Sent Events and bounded LLM cost with softmax-weighted agent selection, concurrent inference, and deterministic state updates.

LaunchPilot | [Devpost](#) · [GitHub](#)

Next.js, TypeScript, FastAPI, Auth0, Backboard.io

- Built an AI launch copilot in 72 hours (2-person team) at Hack Canada 2026; won sole Backboard.io Best Use prize (1 of 59, \$500) and placed top 3 of 94 in the SPUR Build a Real Canadian Startup track (\$500 grant).
- Developed a supervised launch workflow that turns a project brief or GitHub repository into market research, positioning, ICPs, messaging, KPIs, execution plans, and outreach drafts.
- Orchestrated research, positioning, and execution stages of 5 agents each via persistent threads and memory snapshots so each stage reused structured context from prior runs.
- Integrated GitHub context ingestion, Auth0-scoped permissions, and approval-gated execution actions to support secure, repository-aware recommendations.

Market Regime Forecasting | [GitHub](#)

Python, Pandas, Flask, pytest

- Implemented a first-order Markov chain modeling daily equity return regimes over ~60 trading days.
- Cleaned CSV prices, computed returns, and built a row-normalized 3×3 transition matrix for forecasts.
- Built a Flask dashboard with threshold tuning and 10,000+ Monte Carlo simulations for sensitivity analysis.
- Wrote unit tests for state classification and transition logic to validate probability correctness.

UofT Theft Data Map | [Live Demo](#) · [GitHub](#)

Next.js, TypeScript, FastAPI, SQLite

- Built a full-stack app mapping Toronto Police theft-over-\$5,000 incidents around the UofT St. George campus.
- Created a Python/FastAPI pipeline to filter incidents by campus boundaries and normalize records in SQLite.
- Implemented an interactive OpenStreetMap heatmap with summary metrics and hotspot visualization.

Experience

YM Tech Council

Sep 2023 – Jul 2025

- Provided real-time technical support across 3 robotics/programming tournaments per semester (70+ participants each): setup, infrastructure, and troubleshooting.
- Coordinated scheduling, logistics, and resources for multi-team technical and student programming events.